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3D Medical Imaging · Multimodal AI · Trustworthy AI

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Research Profile

I study the reliability and generalization of medical AI beyond aggregate benchmark performance, focusing on **3D medical imaging**, **multimodal image-text modeling**, and **reliability auditing**. I am preparing for **2027 PhD applications** in medical imaging, biomedical engineering, computer science, and related AI fields. Research conducted with the group at Shengjing Hospital, China Medical University.

Education

China Medical University, Shenyang, China 2024.09 – 2027.06

Master of Library and Information Science (MLIS); research focus: Medical AI

Jilin University, Changchun, China 2018.08 – 2022.06

B.Mgmt. in Archival Science; B.Econ. in International Economics and Trade

Research Focus

- **3D medical imaging and generalization:** lung CT classification, longitudinal imaging, registration, calibration, matched architecture controls, external-cohort validation, and source-free adaptation.
- **Multimodal and grounded medical AI:** image-text fusion, clinical-text provenance, guideline-grounded RAG, evidence sufficiency, abstention, and structured decision realization.
- **Trustworthy evaluation:** shared failure patterns, distribution shift, test-time update risk, data provenance, unsupported claims, selective referral, and reproducible audit pipelines.

Selected Research Outputs

Selected to represent my current PhD research direction. † Equal contribution. Double-anonymous submissions are described without venue identifiers.

A Benefit-Risk Audit of Test-Time Update Budgets for Longitudinal Medical Image Registration: A GrowthMamba Case Study

Xin Meng[†], Xiaopeng Li[†], Huixin Yu[†], Qing Chang, Yuhong Zhao

Oral presentation, 32nd Chinese Medical Association Conference on Medical Informatics, 2026

Audited frozen inference against multiple test-time update configurations; under a CT-to-brain-MRI stress test, frozen inference outperformed all three online-update variants in mean similarity while maintaining lower folding risk, motivating reject/rollback rather than default adaptation. **My role:** designed the benefit-risk audit and analyzed similarity and Jacobian-folding risk across the frozen and test-time update configurations.

Source-calibrated CT deep learning for lung nodule malignancy risk: external validation and matched-control analysis across public cohorts

Xin Meng[†], Xiaopeng Li[†], Huixin Yu[†], Yuhong Zhao, Qing Chang

Manuscript under review – Cancer Imaging

Built a source-calibrated 3D CT malignancy-risk workflow and externally validated it on LNDb and LIDC-IDRI; matched controls found no clear performance gain from added Mamba layers or geometric metadata. **My role:** designed the matched CNN/Mamba/attention controls and ran all external-cohort validation and calibration analyses.

Dual-teacher distillation and source-free adaptation for 3D lung-nodule classification

Xin Meng[†], Xiaopeng Li[†], Huixin Yu[†], Yuhong Zhao, Qing Chang

Manuscript under review

Analyzed CNN/Transformer dual-teacher distillation in a SegMamba student: feature-level teacher gradients were consistently aligned whereas logit-level KD often conflicted; external LNDb evaluation showed competitive transfer but no significant TTA gain. **My role:** implemented the dual-teacher SegMamba experiments and ran the gradient-conflict, external-transfer, test-time adaptation, and compute analyses.

Mining Multi-Model Failure Patterns for Reliability Auditing of 3D Lung Nodule Malignancy Predictors

Xin Meng[†], Xiaopeng Li[†], Huixin Yu[†], Yuhong Zhao, Qing Chang

Manuscript under review

Converted per-case outputs from seven 3D models into a reusable behavior table for co-failure and referral auditing; shared errors persisted despite strong ensemble discrimination, while referral analysis exposed class-specific residual risk. **My role:** built the multi-model failure-mining pipeline and the selective-referral analysis.

Selected Research Outputs (continued)

Auditable Decision-Invariant Retrieval-Augmented Lung-RADS Report Realization: A Verified-Subset Technical Evaluation

Xin Meng[†], Xiaopeng Li[†], Huixin Yu[†], Yuhong Zhao, Qing Chang
Manuscript under review – IEEE Journal of Biomedical and Health Informatics

Designed a decision-invariant Lung-RADS RAG pipeline in which deterministic rules own categorization and the LLM only realizes a fixed decision; audited citation validity, abstention, leakage, and unsupported generation. *My role: designed the decision-invariant pipeline and ran the citation, abstention, leakage, and unsupported-generation audits.*

Selected Additional Research

Additional selected work extending the same reliability and grounded-AI themes.

Clinical-text leakage and text-dominance auditing in multimodal lung-nodule assessment. Xin Meng, co-first author (listed first). *Manuscript under review.*

Source-bounded answerability and over-decision benchmarking for grounded RAG. Xin Meng, co-first author (listed first). *Manuscript under review.*

General-purpose VLMs versus task-specific 3D models for pulmonary nodule malignancy prediction. Xin Meng, co-first author (listed first). *Manuscript under review.*

Co-authored (non-first-author) manuscripts under review: clinical-nutrition knowledge graph (AIIM); recipe substitution and nutrient computation (JFCA); cross-cultural diffusion of sustainable recipes (HSSC).

Conference & Honors

- Oral presenter and youth-group science-communication presenter at the 32nd Chinese Medical Association Conference on Medical Informatics (2026).
- First-authored paper accepted, 31st Chinese Medical Association Conference on Medical Informatics (2025).
- Science Communication Pioneer Award, science communication competition (2026).
- Information Systems Supervisor, National Intermediate Professional Certificate.

Research Experience

National Health Commission Health Development Research Center 2025.11 – 2026.02
Remote Research Assistant, Prof. Wudong Guo’s group

- Collected and harmonized multi-province expert consensus documents and group standards on off-label drug use; built a structured comparison table for retrieval and downstream analysis.

Technical Skills

Programming	Python, PyTorch, pandas, SQL, Git/GitHub; experiment automation and reproducible analysis pipelines.
Medical imaging	3D CT preprocessing, HU normalization, voxel-spacing handling, longitudinal pairing, patient-level splitting, external-cohort evaluation.
Modeling	3D CNNs, Mamba/state-space models, Transformers, knowledge distillation, multimodal fusion, RAG, agent workflows.
Evaluation	Patient-clustered bootstrap, calibration, fixed operating points, ablations, matched controls, subgroup/failure analysis, selective referral, provenance audits.
Languages	Chinese (native); English: IELTS 6.5 (Reading 7.5, 2024), CET-6 502.

Prior Data Experience

Shanghai Zhizhongda Information Technology Co., Ltd. – Data Analyst, 2024.05–2024.08. SQL/Tableau analysis of large quotation and order tables, competitor-price monitoring, and pricing-rule support.

Gold Hongye Paper Group (APP / Qingfeng) – Data Analysis Management Trainee, 2023.04–2024.05. Campaign and pricing analytics across five regions and 100+ core SKUs, including JD.com 618 campaign support.